

Notes: Bates, Kahle, and Stulz (JF, 2009)
Why Do U.S. Firms Hold So Much More Cash than They Used To?

Contents

1	Big picture	4
1.1	Question	4
1.2	Setting	4
1.3	Main answer	4
1.4	Core empirical facts	4
1.5	Economic takeaway	5
2	Incremental contribution of the paper	5
2.1	Relative to earlier cash-holding papers	5
2.2	Relative to disappearing-dividend and new-listings papers	5
2.3	Relative to leverage literature	5
2.4	Relative to agency explanations	6
3	Theory: why firms hold cash	6
3.1	Transaction motive	6
3.2	Precautionary motive	6
3.3	Tax motive	6
3.4	Agency motive	7
4	Basic facts: cash, leverage, and net leverage	7
4.1	Cash ratios	7
4.2	Leverage versus net leverage	7
5	Where is the increase concentrated?	7
5.1	Firm size	7
5.2	Newly public firms	8
5.3	Dividend payers and nondividend payers	8
5.4	Firms with losses	8
5.5	Cash-flow volatility	8
5.6	IPO cohorts and the changing listed-firm population	8

5.7	High-tech firms and manufacturing firms	9
5.8	Foreign income and tax effects	9
6	Cash-demand regressions	9
6.1	Baseline empirical model	9
6.2	Expected signs	9
6.3	Baseline regression findings	10
6.4	Changes regressions	10
6.5	Do intercepts shift over time?	10
6.6	Do slopes change over time?	10
7	Why did the cash ratio increase?	10
7.1	Out-of-sample 1980s model	10
7.2	Prediction results	11
7.3	Decomposition of predicted cash growth	11
7.4	Declining working capital	11
7.5	Increasing cash-flow risk	11
7.6	Declining capital expenditures and rising R&D	12
7.7	Dividend payers versus nonpayers	12
7.8	S&P 500 versus non-S&P 500 firms	12
8	Agency tests	12
8.1	Motivation	12
8.2	GIM governance-index test	12
8.3	Value of cash test	12
8.4	Excess cash and future cash growth	13
9	Interpretation and mechanisms	13
9.1	Why precautionary cash rises despite better financial technology	13
9.2	Cash as a buffer for intangible firms	13
9.3	Cash and net working capital as substitutes	14
9.4	Why net debt matters	14
10	Identification, robustness, and limitations	14
10.1	What the paper identifies	14

10.2 Why the evidence is persuasive	14
10.3 Limitations	14
11 Extracted figures and tables from the paper	15
11.1 Table I: Average and Median Cash and Leverage Ratios from 1980 to 2006; Figure 1: Average cash ratios by firm size quintile from 1980 to 2006	15
11.2 Table II: Average Cash Ratios from 1980 to 2006 Delineated by New Issue Status, the Payment of Dividends, and Accounting Performance; Figure 2: Average cash ratios by idiosyncratic risk	15
11.3 Figure 3: Average cash ratios by an IPO cohort; Figure 4: Average cash ratios sorted by quintiles of the GIM index from 1990 to 2006	16
11.4 Table III: Regressions Estimating the Determinants of Cash Holdings, Panel A, part 1; Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel A, part 2	17
11.5 Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel B, part 1; Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel B, part 2	17
11.6 Table IV: Predicted Cash Ratios and Their Deviations from Actual Cash Holdings over Time; Table V: Determinants of Changes in Predicted Cash between 2000 and 2006	18
11.7 Table VI: OLS Regression Results for the Market Value of the Firm	19
12 Short summary	20
13 Conclusion	20

1 Big picture

1.1 Question

The paper asks why U.S. industrial firms hold much more cash than they used to. The empirical fact is large: the average cash-to-assets ratio of U.S. industrial firms more than doubles from 1980 to 2006. The question is whether this increase reflects precautionary cash demand, changing firm characteristics, tax incentives, transaction motives, or agency problems.

The paper also connects the cash increase to capital structure measurement. If cash is economically close to negative debt, then gross debt ratios can hide a large fall in net leverage. By the end of the sample, the average firm can essentially retire all debt obligations with cash holdings.

1.2 Setting

The sample is U.S.-incorporated industrial firms in Compustat from 1980 to 2006. Financial firms and utilities are excluded. The baseline sample contains 117,438 firm-year observations for 13,599 unique firms. The central variable is the cash ratio:

$$\text{Cash ratio}_{it} = \frac{\text{Cash and marketable securities}_{it}}{\text{Book assets}_{it}}. \quad (1)$$

The paper compares this measure with leverage and net leverage:

$$\text{Leverage}_{it} = \frac{\text{Long-term debt}_{it} + \text{Debt in current liabilities}_{it}}{\text{Book assets}_{it}}, \quad (2)$$

$$\text{Net leverage}_{it} = \frac{\text{Debt}_{it} - \text{Cash}_{it}}{\text{Book assets}_{it}}. \quad (3)$$

1.3 Main answer

The evidence points primarily to the precautionary motive. Firms hold more cash because their cash flows become riskier, their working capital becomes less liquid or less available as a cash substitute, their capital expenditures decline, and their R&D intensity rises. These changes increase the value of internal liquidity buffers.

The paper finds little systematic support for the view that the aggregate increase in cash is mainly an agency problem. Cash does not grow most in firms with more entrenched managers, the value of cash does not fall over time, and firms with high excess cash do not subsequently accumulate cash faster.

1.4 Core empirical facts

The paper establishes four headline facts:

- Average cash/assets rises from 10.5% in 1980 to 23.2% in 2006; the median rises from 5.5% to 13.3%.
- Net leverage falls sharply and is negative for the average firm in 2004, 2005, and 2006.

- The increase is strongest among nondividend payers, negative-income firms, small firms, high cash-flow-risk industries, and newer listing cohorts.
- A model estimated on the 1980s explains much of the later increase using observable firm characteristics.

1.5 Economic takeaway

The paper’s central message is that higher corporate cash is not a mysterious hoarding puzzle. The listed-firm population changes toward firms with more volatile cash flows, more intangible investment, fewer collateralizable assets, and lower working capital buffers. These firms rationally hold more precautionary cash. The implication is that gross leverage alone becomes increasingly misleading; net leverage is essential for understanding modern capital structure.

2 Incremental contribution of the paper

2.1 Relative to earlier cash-holding papers

Earlier work, especially Opler, Pinkowitz, Stulz, and Williamson (1999), identifies cross-sectional determinants of corporate cash holdings. Bates, Kahle, and Stulz extend that literature by asking whether those determinants can explain a long-run time-series increase in cash holdings.

The incremental contribution is threefold. First, the paper documents the magnitude and pervasiveness of the secular increase in cash and the associated fall in net debt. Second, it evaluates whether the increase reflects a regime shift in the demand for cash or a movement along an existing demand function caused by changes in firm characteristics. Third, it directly assesses whether agency explanations can explain the aggregate time-series pattern.

2.2 Relative to disappearing-dividend and new-listings papers

The paper links the cash trend to Fama and French’s evidence on disappearing dividends and changing listed-firm characteristics. Firms that do not pay dividends and firms from newer IPO cohorts hold much more cash. The paper therefore treats the cash increase as part of a broader transformation of the public-firm universe: newer listed firms are less mature, more R&D-intensive, more volatile, and less likely to pay dividends.

2.3 Relative to leverage literature

The paper contributes to capital structure research by showing that debt/assets and debt/equity can miss a dramatic change in financial flexibility. Gross leverage does not fall much, but net leverage falls sharply because cash rises. If cash can be used to repay debt or avoid distress, then a gross-debt view overstates financial risk.

2.4 Relative to agency explanations

Agency theory predicts that managers with weak governance and limited payout pressure may accumulate cash. The paper takes that explanation seriously, but provides three tests that work against it: GIM governance-index sorts, valuation regressions for the value of cash, and future cash growth following excess-cash residuals. The contribution is not that agency never matters, but that it does not systematically explain the aggregate increase.

3 Theory: why firms hold cash

3.1 Transaction motive

The transaction motive says firms hold cash because converting noncash financial assets into cash is costly. Classic inventory-style models imply economies of scale: large firms should hold less cash/assets than small firms. This motive predicts a negative relation between firm size and the cash ratio.

In the sample, the size relation is generally negative, but it weakens in the 2000s. Moreover, improvements in financial technology and cash management would, if anything, tend to reduce transaction cash demand over time. Thus transaction motives alone cannot explain the secular increase.

3.2 Precautionary motive

The precautionary motive says firms hold cash to cushion adverse cash-flow shocks when external finance is costly. The value of cash is high when:

- cash flows are volatile;
- external finance is costly or unavailable;
- investment opportunities are valuable;
- assets are intangible or hard to pledge as collateral;
- distress costs are high.

This motive predicts more cash for high-risk firms, high market-to-book firms, nondividend payers, R&D-intensive firms, and firms with low collateralizable assets. The paper's main evidence supports this motive.

3.3 Tax motive

The tax motive is especially relevant for multinational firms. U.S. firms with foreign earnings may keep cash abroad to avoid repatriation taxes. The paper checks this explanation by comparing firms with and without foreign pretax income. Both groups experience a secular cash increase, and there is no strong evidence that foreign-income firms drive the overall trend.

3.4 Agency motive

The agency motive, following Jensen's free-cash-flow logic, predicts that entrenched managers may retain cash rather than distribute it to shareholders. The strongest version predicts high cash accumulation in firms with poor investment opportunities, high free cash flow, and weak governance.

The paper finds that cash rises most in firms that look financially constrained or risky, not in mature low-growth cash cows. This is why the authors conclude that agency conflicts cannot systematically explain the aggregate trend.

4 Basic facts: cash, leverage, and net leverage

4.1 Cash ratios

The average cash ratio more than doubles over the sample. The increase is visible in both means and medians, so it is not only a few very large cash-rich firms. The average cash ratio increases from 10.5% to 23.2%, and the median from 5.5% to 13.3%.

A simple time trend regression of average cash/assets on year produces a positive and significant slope. The reported slope is about 0.46 percentage points per year for the average cash ratio and about 0.27 percentage points per year for the median.

4.2 Leverage versus net leverage

Gross leverage is relatively stable compared with net leverage. The key accounting distinction is:

$$\text{Net debt ratio}_{it} = \text{Debt ratio}_{it} - \text{Cash ratio}_{it}. \quad (4)$$

Because the cash ratio rises so much, net debt falls even if gross debt does not. The average net debt ratio falls from 16.4% in 1980 to negative values in the mid-2000s.

Economic message. A firm with debt/assets of 20% and cash/assets of 20% has very different effective leverage from a firm with debt/assets of 20% and cash/assets of 2%. The rise in cash means that capital structure research should distinguish gross debt from net debt.

5 Where is the increase concentrated?

5.1 Firm size

Cash ratios rise in every firm-size quintile. The increase is most pronounced among smaller firms, but even the largest firms experience a positive trend. This rules out a simple explanation based only on the cash hoards of large firms such as Microsoft or Exxon.

The size results are consistent with a precautionary interpretation because small firms generally face higher financing frictions and more volatile cash flows. But the fact that cash rises broadly suggests a widespread change in the public-firm environment.

5.2 Newly public firms

IPO firms hold much more cash than older firms. Firms that went public within the prior five years have higher cash ratios than non-IPO firms throughout the sample. However, even non-IPO firms experience a large increase in cash, so the trend is not mechanically caused by IPO proceeds.

The cohort evidence is especially important. Later IPO cohorts hold more cash even after the first five post-listing years. This means newly listed firms are fundamentally different from older cohorts: they are more cash-intensive and likely more risky or R&D-oriented.

5.3 Dividend payers and nondividend payers

The cash increase is concentrated among firms that do not pay dividends. Dividend payers do not show the same strong trend. At the beginning of the sample, dividend payers and nonpayers have more similar cash ratios; by the end, nonpayers hold much more cash.

This is consistent with precautionary cash demand because nondividend payers are often younger, riskier, and more financially constrained. It is also consistent with the disappearing-dividends literature: as listed firms become less likely to pay dividends, their cash policies also become more precautionary.

5.4 Firms with losses

Firms with negative net income experience a dramatic increase in cash holdings. Their average cash ratio almost triples. Firms with nonnegative net income also increase cash, but much less. This points toward cash being accumulated by firms with weak operating performance and higher financing risk, not only by profitable firms with excess free cash flow.

5.5 Cash-flow volatility

The paper sorts industries by the increase in cash-flow volatility. The cash ratio rises most in industries with the greatest increase in cash-flow risk. Firms in the top volatility-increase quintile have an average cash ratio that rises from roughly 12.9% in 1980 to 39.0% in 2006.

Economic message. This is one of the cleanest pieces of evidence for the precautionary motive. If the business environment becomes riskier, and external finance is imperfect, firms optimally hold more liquidity.

5.6 IPO cohorts and the changing listed-firm population

The paper constructs listing cohorts. Later cohorts hold more cash at similar lifecycle stages than earlier cohorts. This fact links the cash trend to the changing nature of public firms. The market increasingly contains firms with weak track records, more intangible investment, and more idiosyncratic risk.

5.7 High-tech firms and manufacturing firms

High-tech firms hold more cash than manufacturing firms, but cash ratios increase in both groups. Thus the rise in cash is not merely a high-tech composition effect. The broad pattern is more general: firms across sectors become more cash-intensive.

5.8 Foreign income and tax effects

Firms with foreign pretax income have incentives to retain cash abroad, but the paper finds that firms without foreign taxable income also experience a strong cash increase. The tax motive may matter for some firms, but it is not the dominant explanation for the aggregate pattern.

6 Cash-demand regressions

6.1 Baseline empirical model

The paper estimates cash-holding regressions based on the OPSW framework. A simplified form is:

$$\begin{aligned} \frac{\text{Cash}_{it}}{\text{Assets}_{it}} = & \alpha + \beta_1 \text{Industry sigma}_{it} + \beta_2 \text{Market-to-book}_{it} + \beta_3 \log(\text{Assets}_{it}) \\ & + \beta_4 \frac{\text{Cash flow}_{it}}{\text{Assets}_{it}} + \beta_5 \frac{\text{NWC}_{it} - \text{Cash}_{it}}{\text{Assets}_{it}} + \beta_6 \frac{\text{Capex}_{it}}{\text{Assets}_{it}} \\ & + \beta_7 \text{Leverage}_{it} + \beta_8 \frac{\text{R\&D}_{it}}{\text{Sales}_{it}} + \beta_9 \text{Dividend dummy}_{it} \\ & + \beta_{10} \frac{\text{Acquisitions}_{it}}{\text{Assets}_{it}} + \varepsilon_{it}. \end{aligned} \tag{5}$$

6.2 Expected signs

The signs map to the theories:

- Industry sigma should be positive under precautionary demand.
- Market-to-book should be positive if growth opportunities make financial distress costly.
- Size should be negative because of transaction-cost scale economies and easier access to capital markets.
- Net working capital should be negative because noncash working capital substitutes for cash.
- Capital expenditures can be negative because capex uses cash and creates collateralizable assets.
- Leverage is typically negative if firms use cash to reduce debt or if debt and cash are substitutes.
- R&D should be positive because intangible investment is hard to finance externally.
- Dividend dummy should be negative if dividend payers are safer and less constrained.

6.3 Baseline regression findings

The full-sample OLS regression shows that cash holdings are positively related to industry cash-flow risk, market-to-book, and R&D. Cash is negatively related to size, net working capital, capex, leverage, dividend paying, and acquisition activity. These signs are broadly consistent with transaction and precautionary motives.

6.4 Changes regressions

The paper also estimates changes regressions. The changes specification asks whether changes in firm characteristics explain changes in cash, controlling for lagged changes and lagged levels. This helps absorb time-invariant firm heterogeneity.

The results remain broadly supportive of the main story, although some coefficients differ in sign and significance. The paper does not rely on one specification; instead it checks whether the same economic forces appear across models.

6.5 Do intercepts shift over time?

To test whether cash demand shifts upward for reasons unrelated to measured characteristics, the authors add 1990s and 2000s dummy variables. In the cash/assets specification, the intercepts are negative, not positive. This means that measured firm characteristics predict even higher cash ratios than firms actually hold in those periods.

Some specifications using log cash/net assets suggest a 2000s intercept increase, but this result disappears when slope changes are allowed. Overall, the paper concludes that there is no robust unexplained upward shift in cash demand.

6.6 Do slopes change over time?

The paper allows coefficient slopes to differ in the 1990s and 2000s. The explanatory power barely increases, meaning that a stable model already captures much of the variation. Some slopes change; notably, the negative relation between cash and size weakens in the 2000s. But the main conclusion remains that firm characteristics explain much of the trend.

7 Why did the cash ratio increase?

7.1 Out-of-sample 1980s model

The authors estimate a cash model using only 1980s data and then use it to predict cash ratios in the 1990s and 2000s. This is a demanding test because it asks whether an earlier-period model can explain later-period cash levels.

The model is:

$$\widehat{\text{Cash ratio}}_{it} = \hat{\alpha}_{1980s} + \sum_k \hat{\beta}_{k,1980s} X_{kit}. \quad (6)$$

The prediction error is:

$$\text{Prediction error}_{it} = \text{extActualcashratio}_{it} - \widehat{\text{Cash ratio}}_{it}. \quad (7)$$

7.2 Prediction results

The 1980s model overpredicts cash in much of the 1990s and only modestly underpredicts cash near the end of the sample. For the whole sample, the model explains a large share of the increase. For dividend payers, the model generally predicts more cash than they actually hold. For nondividend payers, prediction errors become positive in the later sample, but the errors are not large enough to overturn the main explanation.

7.3 Decomposition of predicted cash growth

The predicted change in average cash can be decomposed as:

$$\Delta \widehat{\text{Cash ratio}} = \sum_k \hat{\beta}_{k,1980s} \Delta \bar{X}_k. \quad (8)$$

This equation attributes the predicted cash increase to changes in average firm characteristics.

The four most important contributors are:

1. declining noncash net working capital;
2. rising cash-flow volatility;
3. declining capital expenditures;
4. rising R&D intensity.

7.4 Declining working capital

Net working capital net of cash falls substantially. Inventories and accounts receivable decline as a fraction of assets. Since inventories and receivables can be converted into cash or used as operating liquidity buffers, their decline increases the need for cash balances.

In the model, the coefficient on NWC/assets is negative. Therefore, when NWC/assets falls, predicted cash/assets rises:

$$\beta_{NWC} < 0, \quad \Delta NWC < 0 \quad \Rightarrow \quad \beta_{NWC} \Delta NWC > 0. \quad (9)$$

7.5 Increasing cash-flow risk

Average industry cash-flow risk more than doubles from the 1980s to the 2000s. Since the coefficient on industry sigma is positive, this mechanically raises predicted cash:

$$\beta_{\sigma} > 0, \quad \Delta \sigma > 0 \quad \Rightarrow \quad \beta_{\sigma} \Delta \sigma > 0. \quad (10)$$

7.6 Declining capital expenditures and rising R&D

Capital expenditures decline, while R&D rises. This is a central structural change. Capital expenditures create tangible assets that can be pledged as collateral, while R&D creates intangible assets that are harder to finance externally and more vulnerable to financial distress.

Because cash is negatively related to capex and positively related to R&D, the shift from capex toward R&D raises cash demand. This is a key intuition: the modern firm has fewer tangible assets and more intangible growth options, so internal liquidity becomes more valuable.

7.7 Dividend payers versus nonpayers

The decomposition helps explain why nondividend payers drive the aggregate trend. Nondividend payers experience greater increases in risk, more R&D intensity, and leverage changes that raise predicted cash. Dividend payers also experience changes in NWC and risk, but the total predicted increase is smaller.

7.8 S&P 500 versus non-S&P 500 firms

Predicted cash grows more for non-S&P 500 firms than for S&P 500 firms. The difference is mainly due to R&D and leverage. Non-S&P 500 firms become more R&D-intensive and have larger changes in characteristics associated with precautionary cash demand.

8 Agency tests

8.1 Motivation

Agency theory could explain cash accumulation if entrenched managers prefer to keep resources under their control rather than return cash to shareholders. The paper tests this explanation directly rather than simply assuming precautionary motives.

8.2 GIM governance-index test

The paper uses the Gompers, Ishii, and Metrick governance index as a proxy for managerial entrenchment. Higher GIM values indicate more antitakeover provisions and more entrenched management.

If agency problems explain the cash rise, firms with high GIM scores should experience the largest cash increases. The paper finds the opposite pattern: firms in the highest GIM quintile do not have the largest cash increases and often have lower cash ratios than firms with low GIM scores.

8.3 Value of cash test

The paper estimates valuation regressions based on Fama and French (1998) and Pinkowitz and Williamson (2004). The model relates the market value of the firm to earnings, changes in earnings,

net assets, R&D, interest expense, dividends, future value changes, and cash:

$$\begin{aligned}
V_{it} = & \alpha + \beta_1 E_{it} + \beta_2 dE_{it} + \beta_3 dE_{i,t+2} + \beta_4 dNA_{it} + \beta_5 dNA_{i,t+2} \\
& + \beta_6 RD_{it} + \beta_7 dRD_{it} + \beta_8 dRD_{i,t+2} + \beta_9 I_{it} + \beta_{10} dI_{it} \\
& + \beta_{11} dI_{i,t+2} + \beta_{12} D_{it} + \beta_{13} dD_{it} + \beta_{14} dD_{i,t+2} \\
& + \beta_{15} dV_{i,t+2} + \beta_{16} L_{it} + \varepsilon_{it},
\end{aligned} \tag{11}$$

where L_{it} is cash holdings and NA is net assets excluding cash.

To test whether cash becomes less valuable, the paper interacts cash with 1990s and 2000s dummies. If agency problems increasingly explain cash accumulation, the marginal value of cash should fall. The paper finds no evidence of a decline in the value of cash.

8.4 Excess cash and future cash growth

The final agency test ranks firms by excess cash, defined as residual cash from the cash-holding model. If agency problems cause firms to hoard cash, high-excess-cash firms should subsequently accumulate cash faster. Instead, excess cash is negatively related to future cash growth.

Economic message. Agency conflicts may explain cash holdings for some firms, but they do not explain the aggregate increase. The firms that increase cash most look risky and constrained, and the value of cash does not fall.

9 Interpretation and mechanisms

9.1 Why precautionary cash rises despite better financial technology

A puzzle is that derivatives, banking technology, and cash management improved over the sample. These developments should reduce the need for cash. The paper resolves this by emphasizing that firms also become riskier and more intangible. Improved financial technology may reduce transaction balances, but it does not fully insure firms against unhedgeable operating risk, external finance costs, or intangible-investment financing frictions.

9.2 Cash as a buffer for intangible firms

R&D-intensive firms cannot easily pledge their investment opportunities as collateral. If a negative cash-flow shock occurs, such firms may be forced to cut valuable R&D or issue costly external capital. Cash helps avoid underinvestment.

This mechanism makes the precautionary motive particularly strong for firms with:

- high R&D;
- high growth opportunities;
- low tangible assets;
- volatile cash flows;

- no dividend payout.

9.3 Cash and net working capital as substitutes

Noncash working capital can provide liquidity. Inventories can be liquidated, receivables can be collected or securitized, and working capital can be managed to free cash. As firms hold fewer inventories and receivables, cash becomes a more important liquidity buffer.

9.4 Why net debt matters

The paper argues that modern capital structure analysis should pay attention to net debt. The same gross leverage ratio means less financial fragility when cash balances are large. Since average net debt becomes negative near the end of the sample, conclusions about leverage trends depend heavily on whether cash is netted out.

10 Identification, robustness, and limitations

10.1 What the paper identifies

The paper primarily identifies whether observable characteristics known to predict cash holdings can account for the time-series increase in cash. It does not rely on a single natural experiment. Instead, it uses many cross-sectional and time-series patterns to assess which motives are consistent with the data.

10.2 Why the evidence is persuasive

The evidence is persuasive because multiple independent facts point in the same direction. Cash rises more among risky, young, nondividend-paying, negative-income, and high-volatility firms. The regression model links cash to risk, R&D, NWC, capex, and leverage. The 1980s model predicts much of the later cash increase. Agency proxies do not line up with the trend.

10.3 Limitations

The cash-holding regressions are not structural estimates of a dynamic cash policy model. Some explanatory variables, such as leverage, NWC, and cash, are jointly determined. The paper acknowledges that the empirical cash-demand model is based on the existing literature and is not a definitive theory. Still, the robustness across specifications makes the qualitative conclusions strong.

The sample ends in 2006, before the global financial crisis. Later periods may feature additional forces, such as post-crisis regulation, zero interest rates, tax reform, and changing payout policy. The paper is best read as an explanation of the 1980-2006 secular rise.

11 Extracted figures and tables from the paper

11.1 Table I: Average and Median Cash and Leverage Ratios from 1980 to 2006; Figure 1: Average cash ratios by firm size quintile from 1980 to 2006

Table I
Average and Median Cash and Leverage Ratios from 1980 to 2006
The sample includes all Compustat firm-year observations from 1980 to 2006 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 117,438 observations for 13,599 unique firms. Variable definitions are provided in the Appendix.

Year	N	Aggregate Cash Ratio	Average Cash Ratio	Median Cash Ratio	Average Leverage	Median Leverage	Average Net Leverage	Median Net Leverage
1980	3,519	0.063	0.105	0.055	0.269	0.243	0.164	0.178
1981	3,748	0.057	0.121	0.058	0.253	0.228	0.133	0.160
1982	3,752	0.061	0.121	0.064	0.261	0.232	0.140	0.158
1983	4,120	0.076	0.159	0.087	0.246	0.204	0.087	0.111
1984	4,172	0.070	0.140	0.069	0.254	0.218	0.114	0.141
1985	4,127	0.069	0.142	0.070	0.270	0.230	0.128	0.151
1986	4,261	0.076	0.157	0.081	0.273	0.236	0.116	0.143
1987	4,407	0.077	0.156	0.077	0.273	0.241	0.116	0.153
1988	4,237	0.062	0.141	0.068	0.280	0.244	0.139	0.163
1989	4,095	0.055	0.138	0.063	0.286	0.253	0.148	0.173
1990	4,042	0.051	0.134	0.062	0.282	0.244	0.147	0.168
1991	4,137	0.055	0.155	0.072	0.259	0.215	0.104	0.129
1992	4,307	0.057	0.163	0.079	0.245	0.193	0.082	0.111
1993	4,713	0.060	0.171	0.083	0.225	0.179	0.053	0.091
1994	4,985	0.058	0.155	0.070	0.230	0.187	0.075	0.106
1995	5,165	0.060	0.171	0.072	0.230	0.187	0.059	0.105
1996	5,568	0.066	0.193	0.088	0.222	0.170	0.029	0.077
1997	5,605	0.068	0.191	0.089	0.236	0.180	0.046	0.085
1998	5,263	0.065	0.178	0.075	0.289	0.205	0.110	0.119
1999	4,971	0.075	0.194	0.077	0.247	0.198	0.053	0.104
2000	4,947	0.074	0.208	0.088	0.242	0.173	0.034	0.075
2001	4,540	0.080	0.214	0.107	0.268	0.173	0.054	0.062
2002	4,233	0.091	0.214	0.114	0.258	0.172	0.045	0.054
2003	3,992	0.101	0.227	0.133	0.235	0.160	0.008	0.016
2004	3,893	0.109	0.240	0.147	0.225	0.145	-0.015	-0.003
2005	3,549	0.105	0.237	0.148	0.219	0.136	-0.020	-0.005
2006	3,297	0.103	0.232	0.133	0.221	0.146	-0.010	0.015

Table I: Average and Median Cash and Leverage Ratios from 1980 to 2006

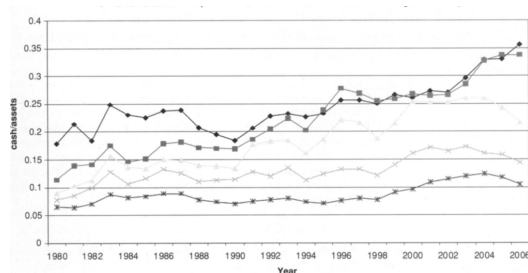


Figure 1. Average cash ratios by firm size quintile from 1980 to 2006. The sample includes all Compustat firm-year observations from 1980 to 2006 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are also excluded from the sample, yielding a panel of 117,438 observations for 13,599 unique firms. The cash ratio is measured as the ratio of cash and marketable securities to the book value of total assets. Firms are sorted into quintiles based on the book value of sample firm assets in the prior fiscal year. The first quintile (Q1) comprises the smallest firms in the sample, while the fifth quintile (Q5) comprises the largest firms in the sample.

Figure 1: Average cash ratios by firm size quintile from 1980 to 2006

Read.

- Table I documents the core empirical fact: average and median cash ratios rise strongly from 1980 to 2006.
- Average net leverage falls from 16.4% in 1980 to negative values in 2004-2006.
- Figure 1 shows the cash-ratio increase across all size quintiles, not just among the largest firms.

Economic message. The cash increase is broad, and its capital-structure implication is large: net leverage falls much more than gross leverage.

11.2 Table II: Average Cash Ratios from 1980 to 2006 Delineated by New Issue Status, the Payment of Dividends, and Accounting Performance; Figure 2: Average cash ratios by idiosyncratic risk

Read.

- Table II shows that IPO firms, nondividend payers, and negative-income firms hold substantially more cash.
- The increase for nondividend payers is much larger than for dividend payers.

Table II
Average Cash Ratios from 1980 to 2006 Delineated by New Issue Status, the Payment of Dividends, and Accounting Performance

The sample includes all Compustat firm-year observations from 1980 to 2006 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 117,438 observations for 13,599 unique firms. Firms are assigned to the IPO subsample if they have gone public within the prior 5 calendar years, and to the non-IPO subsample otherwise. A firm is classified as a dividend payer if it paid common dividends in that year. Firms with accounting losses at the fiscal end of the designated year are assigned to the negative net income subsample. Differences in the average cash ratio between the new issues, dividend status, and accounting performance subsamples are statistically different from zero at better than the 1% level for each reported year with the exception of differences in accounting performance for 1982. Variable definitions are provided in the Appendix.

Year	New Issues		Dividend Status		Accounting Performance	
	IPO Firms	Non-IPO Firms	Dividend Payer	Nondividend Payer	Negative Net Income	Nonnegative Net Income
1980	0.211	0.099	0.086	0.130	0.122	0.101
1981	0.231	0.109	0.092	0.151	0.140	0.115
1982	0.209	0.110	0.103	0.138	0.119	0.122
1983	0.275	0.131	0.118	0.189	0.173	0.153
1984	0.214	0.117	0.101	0.165	0.159	0.132
1985	0.206	0.120	0.106	0.164	0.150	0.138
1986	0.225	0.132	0.111	0.181	0.169	0.151
1987	0.209	0.134	0.109	0.178	0.182	0.143
1988	0.187	0.126	0.103	0.159	0.165	0.129
1989	0.181	0.125	0.098	0.156	0.147	0.132
1990	0.187	0.120	0.097	0.151	0.145	0.128
1991	0.245	0.132	0.103	0.177	0.172	0.144
1992	0.262	0.135	0.104	0.188	0.193	0.146
1993	0.265	0.136	0.105	0.198	0.214	0.148
1994	0.222	0.125	0.092	0.179	0.206	0.132
1995	0.248	0.131	0.096	0.198	0.207	0.152
1996	0.276	0.143	0.097	0.224	0.262	0.154
1997	0.263	0.149	0.102	0.218	0.252	0.154
1998	0.251	0.143	0.088	0.205	0.235	0.140
1999	0.302	0.146	0.084	0.225	0.265	0.145
2000	0.327	0.157	0.079	0.239	0.280	0.144
2001	0.345	0.175	0.090	0.243	0.273	0.147
2002	0.362	0.180	0.099	0.241	0.266	0.160
2003	0.380	0.200	0.126	0.256	0.289	0.182
2004	0.402	0.217	0.131	0.276	0.337	0.189
2005	0.324	0.227	0.131	0.276	0.332	0.188
2006	0.326	0.218	0.120	0.277	0.351	0.176

Table II: Average Cash Ratios from 1980 to 2006 Delineated by New Issue Status, the Payment of Dividends, and Accounting Performance

- Figure 2 shows that industries with the largest increase in cash-flow volatility experience the largest rise in cash ratios.

Economic message. These patterns point to precautionary demand. Cash rises most among firms that appear riskier or more financially constrained.

11.3 Figure 3: Average cash ratios by an IPO cohort; Figure 4: Average cash ratios sorted by quintiles of the GIM index from 1990 to 2006

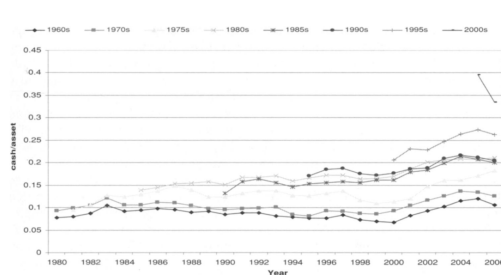


Figure 3. Average cash ratios by an IPO cohort. The figure summarizes the average cash-to-assets ratio for cohorts of firms constructed by listing date. The 1960s cohort includes all firms that have a listing prior to 1970. The 1970s cohort includes all firms that list from 1970 to 1975. We then construct cohorts of firms that list within a 5-year period. Cash holdings for each firm in each cohort are estimated beginning in the 6th year after the listing date. The sample includes all Compustat firm-year observations from 1980 to 2006 with nonmissing data for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 117,438 observations for 13,599 unique firms. The cash ratio is measured as the ratio of cash and marketable securities to the book value of total assets.

Figure 3: Average cash ratios by an IPO cohort

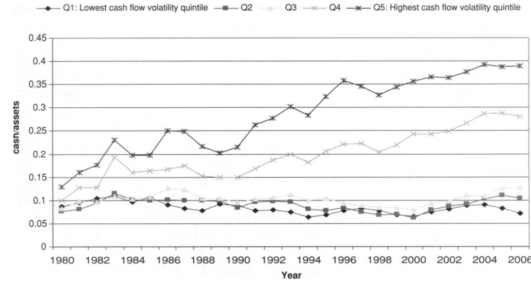


Figure 2. Average cash ratios by idiosyncratic risk. The figure summarizes the average cash-to-assets ratio for quintiles of industries sorted by increases in idiosyncratic risk. We first divide the two-digit SIC code industries in our sample into industry quintiles according to the increase in idiosyncratic cash flow volatility over our sample period. We measure cash flow risk as the standard deviation of industry cash flow computed as follows. For each firm, we compute cash flow standard deviation for the previous 10 years. We require at least three observations for the standard deviation to be calculated. We then take the average across the two-digit SIC code of the firm cash flow standard deviations. The sample includes all Compustat firm-year observations from 1980 to 2006 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 117,438 observations for 13,599 unique firms. The cash ratio is measured as the ratio of cash and marketable securities to the book value of total assets.

Figure 2: Average cash ratios by idiosyncratic risk

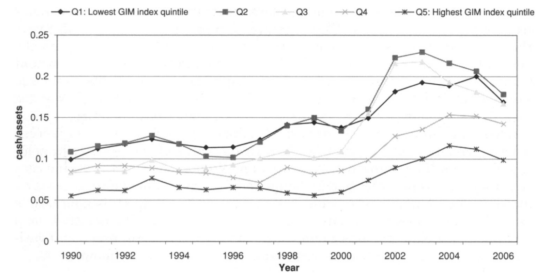


Figure 4. Average cash ratios sorted by quintiles of the GIM index from 1990 to 2006. The figure summarizes the average cash-to-assets ratio for quintiles of firms sorted by their GIM index. We split the firms for which we have the GIM index into quintiles in each year that IRCC publishes its information on takeover protections, using values of the GIM index as break points. We keep the assignment of firms the same in years in which no information is published by IRCC. The sample includes all Compustat firm-year observations from 1990 to 2006 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States.

Figure 4: Average cash ratios sorted by quintiles of the GIM index from 1990 to 2006

Read.

- Figure 3 shows that later IPO cohorts hold more cash than earlier cohorts at similar lifecycle stages.
- Figure 4 shows that cash does not grow most for firms with the highest GIM entrenchment scores.

Economic message. The cohort evidence supports a changing-firm-composition story. The GIM figure works against a simple agency-hoarding explanation.

11.4 Table III: Regressions Estimating the Determinants of Cash Holdings, Panel A, part 1; Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel A, part 2

Table III Regressions Estimating the Determinants of Cash Holdings									
The sample includes all Compustat firm-year observations from 1980 to 2006 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 117,438 observations for 11,599 unique firms. Missing explanatory values reduce the panel used here to 100,414 firm-year observations for 12,792 unique firms for the OLS regressions. The calculation of differences further reduces the sample for the changes regressions to 75,064 firm-year observations. The fixed effects regression excludes firms that have gone public in the past 5 years and has 65,183 firm-year observations. The models in Panel B include separate slopes and intercepts for firm-year observations from the 1990s to the end of the sample period, and for 2000 through 2006. <i>p</i> -values based on standard errors robust to clustering by firm and year are reported in parentheses for the OLS regressions. <i>p</i> -values in the Fama-MacBeth regressions use Newey and West (1987) standard errors to control for autocorrelation. For the fixed effects regression in column 9, we report within <i>R</i> ² . Variable definitions are provided in the Appendix.									
Panel A									
Model	1	2	3	4	5	6	7	8	9
Dependent Variable	OLS	OLS	Changes	OLS	OLS	Changes	F-M (1980s)	F-M (1990s)	FE
	Cash/Assets	Log (Cash/Net Assets)	Cash/Assets	Cash/Assets	Log (Cash/Net Assets)	Cash/Assets	Cash/Assets	Cash/Assets	Cash/Assets
Intercept	0.238 (0.000)	-2.305 (0.000)	0.014 (0.000)	0.261 (0.000)	-2.238 (0.000)	0.010 (0.000)	0.301 (0.000)	0.225 (0.000)	
Lag cash			-0.135 (0.000)		-0.140 (0.000)				
Industry sigma	0.370 (0.000)	0.455 (0.000)	0.578 (0.000)	0.481 (0.000)	0.427 (0.000)	0.578 (0.000)	0.193 (0.000)	0.475 (0.000)	0.090 (0.016)
Market to book	0.016 (0.000)	0.173 (0.000)	0.003 (0.000)	0.016 (0.000)	0.177 (0.000)	0.003 (0.000)	0.011 (0.000)	0.018 (0.000)	0.008 (0.000)
Real size	-0.006 (0.000)	-0.065 (0.000)	0.033 (0.000)	-0.004 (0.000)	-0.068 (0.000)	0.033 (0.000)	-0.009 (0.012)	-0.003 (0.055)	0.004 (0.000)
Cash flow/assets	-0.001 (0.954)	-0.334 (0.000)	0.035 (0.000)	-0.002 (0.863)	-0.328 (0.000)	0.034 (0.000)	0.023 (0.067)	-0.004 (0.768)	0.029 (0.000)

(continued)

Table III—Continued									
Panel A									
Model	1	2	3	4	5	6	7	8	9
Dependent Variable	OLS	OLS	Changes	OLS	OLS	Changes	F-M (1980s)	F-M (1990s)	FE
	Cash/Assets	Log (Cash/Net Assets)	Cash/Assets	Cash/Assets	Log (Cash/Net Assets)	Cash/Assets	Cash/Assets	Cash/Assets	Cash/Assets
NWC/assets	-0.195 (0.000)	-0.552 (0.000)	-0.140 (0.000)	-0.200 (0.000)	-0.533 (0.000)	-0.139 (0.000)	-0.206 (0.000)	-0.202 (0.000)	-0.180 (0.000)
Capex	-0.291 (0.000)	1.353 (0.000)	-0.179 (0.000)	-0.308 (0.000)	1.380 (0.000)	-0.179 (0.000)	-0.263 (0.000)	-0.380 (0.000)	-0.243 (0.000)
Leverage	-0.364 (0.000)	-1.235 (0.000)	-0.188 (0.000)	-0.366 (0.000)	-1.195 (0.000)	-0.186 (0.000)	-0.364 (0.000)	-0.261 (0.000)	-0.234 (0.000)
R&D sales	0.006 (0.000)	0.305 (0.000)	0.010 (0.000)	0.005 (0.000)	0.308 (0.000)	0.010 (0.000)	0.009 (0.000)	0.064 (0.000)	0.019 (0.000)
Dividend dummy	-0.040 (0.000)	-0.270 (0.000)	0.006 (0.960)	-0.045 (0.000)	-0.274 (0.000)	0.005 (0.000)	-0.032 (0.000)	-0.054 (0.000)	0.001 (0.883)
Acquisition activity	-0.171 (0.000)	-0.486 (0.000)	-0.133 (0.001)	-0.170 (0.000)	-0.446 (0.000)	-0.132 (0.002)	-0.093 (0.000)	-0.215 (0.000)	-0.130 (0.000)
1990s dummy						-0.028 (0.000)	-0.175 (0.000)		
2000s dummy						-0.008 (0.000)	0.291 (0.000)	0.012 (0.000)	
Year dummies									Yes
Adjusted R ²	0.455	0.248	0.199	0.459	0.251	0.203	0.357	0.488	0.154

Table III: Regressions Estimating the Determinants of Cash Holdings, Panel A, part 1

Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel A, part 2

Read.

- Panel A presents the main cash-holding regressions using cash/assets and log cash/net assets.
- Industry cash-flow risk, market-to-book, and R&D are generally positively related to cash.
- Size, NWC, leverage, dividends, acquisitions, and often capex are negatively related to cash.

Economic message. The regression signs are consistent with a precautionary and transaction-based demand for cash.

11.5 Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel B, part 1; Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel B, part 2

Read.

- Panel B allows intercepts and slopes to change in the 1990s and 2000s.

Panel B									
Model	1			2			3		
	Cash/Assets			Cash/Assets			Log (Cash/Net Assets)		
Dependent Variable	Estimate	Interaction 1990s	Interaction 2000s	Estimate	Interaction 1990s	Interaction 2000s	Estimate	Interaction 1990s	Interaction 2000s
Intercept	0.306 (0.000)	-0.073 (0.000)	-0.034 (0.010)	0.284 (0.000)	-0.068 (0.000)	-0.016 (0.203)	-1.990 (0.000)	-0.092 (0.378)	-0.251 (0.058)
Industry sigmas	0.142 (0.057)	0.340 (0.000)	0.022 (0.721)	0.222 (0.000)	0.272 (0.000)	0.019 (0.753)	0.151 (0.004)	0.238 (0.004)	0.127 (0.056)
Market to book	0.011 (0.000)	0.007 (0.000)	0.000 (0.983)	0.005 (0.000)	0.006 (0.000)	0.004 (0.007)	0.088 (0.000)	0.084 (0.000)	0.020 (0.239)
Real size	-0.009 (0.000)	0.005 (0.000)	0.005 (0.009)	-0.009 (0.000)	0.005 (0.000)	0.003 (0.030)	-0.076 (0.015)	-0.054 (0.000)	0.062 (0.000)
Cash flow/ assets	0.024 (0.062)	-0.018 (0.272)	-0.041 (0.025)	0.066 (0.000)	0.014 (0.499)	-0.073 (0.002)	-0.177 (0.000)	0.084 (0.245)	-0.057 (0.369)
NWC/ assets	-0.204 (0.000)	-0.006 (0.460)	0.024 (0.048)	-0.226 (0.000)	0.016 (0.330)	0.040 (0.024)	-0.466 (0.000)	-0.134 (0.015)	0.134 (0.277)
Capex	-0.255 (0.000)	-0.071 (0.004)	-0.079 (0.123)	-0.378 (0.000)	-0.067 (0.044)	-0.043 (0.407)	0.070 (0.000)	0.338 (0.000)	-0.048 (0.852)
Leverage	-0.370 (0.000)	0.001 (0.965)	0.013 (0.449)	-0.370 (0.000)	0.001 (0.965)	0.013 (0.449)	-1.443 (0.000)	-0.164 (0.203)	0.527 (0.000)
R&D/sales	0.061 (0.000)	0.003 (0.585)	0.001 (0.834)	0.052 (0.000)	0.004 (0.364)	0.005 (0.083)	0.277 (0.000)	-0.018 (0.523)	0.328 (0.593)
Dividend dummy	-0.055 (0.000)	-0.013 (0.016)	-0.015 (0.007)	-0.016 (0.000)	-0.016 (0.003)	-0.023 (0.000)	-0.054 (0.413)	-0.230 (0.001)	-0.271 (0.000)

(continued)

Table III—Continued									
Panel B									
Model	1			2			3		
	Cash/Assets			Cash/Assets			Log (Cash/Net Assets)		
Dependent Variable	Estimate	Interaction 1990s	Interaction 2000s	Estimate	Interaction 1990s	Interaction 2000s	Estimate	Interaction 1990s	Interaction 2000s
Acquisition activity	-0.076 (0.000)	-0.105 (0.000)	-0.081 (0.000)	-0.219 (0.000)	-0.149 (0.000)	0.020 (0.572)	-0.235 (0.001)	-0.174 (0.000)	0.546 (0.050)
Net equity issuance	0.129 (0.000)	0.022 (0.252)	-0.080 (0.050)	0.260 (0.000)	0.000 (0.967)	0.000 (1.000)	-0.518 (0.000)	-0.518 (0.000)	-0.518 (0.000)
Net debt issuance	0.173 (0.000)	0.023 (0.440)	-0.070 (0.028)	0.298 (0.000)	-0.023 (0.898)	-0.360 (0.000)	-0.298 (0.000)	-0.298 (0.000)	-0.298 (0.000)
Loss dummy	0.002 (0.776)	0.002 (0.776)	0.002 (0.776)	0.002 (0.776)	0.002 (0.776)	0.002 (0.776)	-0.213 (0.001)	-0.213 (0.001)	-0.213 (0.001)
T-bill	0.009 (0.087)	0.009 (0.087)	0.009 (0.087)	0.009 (0.087)	0.009 (0.087)	0.009 (0.087)	0.145 (0.056)	0.145 (0.056)	0.145 (0.056)
Credit spread	0.080 (0.000)	0.080 (0.000)	0.080 (0.000)	0.080 (0.000)	0.080 (0.000)	0.080 (0.000)	0.328 (0.000)	0.328 (0.000)	0.328 (0.000)
IPO1	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.246 (0.000)	0.246 (0.000)	0.246 (0.000)
IPO2	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.171 (0.000)	0.171 (0.000)	0.171 (0.000)
IPO3	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.171 (0.000)	0.171 (0.000)	0.171 (0.000)
IPO4	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.171 (0.000)	0.171 (0.000)	0.171 (0.000)
IPO5	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.171 (0.000)	0.171 (0.000)	0.171 (0.000)
Adjusted R ²	0.462	0.462	0.462	0.462	0.462	0.462	0.274	0.274	0.274

Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel B, part 1

Table III continued: Regressions Estimating the Determinants of Cash Holdings, Panel B, part 2

- Allowing slope changes adds little explanatory power.
- The 2000s do not show a robust unexplained positive intercept shift once slope changes are allowed.

Economic message. The rise in cash is mostly a movement along a cash-demand relation driven by changed firm characteristics, not a pure unexplained regime shift.

11.6 Table IV: Predicted Cash Ratios and Their Deviations from Actual Cash Holdings over Time; Table V: Determinants of Changes in Predicted Cash between 2000 and 2006

Table IV Predicted Cash Ratios and Their Deviations from Actual Cash Holdings over Time									
This table summarizes the predicted cash ratios of sample firms from 1990 through 2006, and deviations of the actual cash ratios from those predicted by an out-of-sample model. Predicted cash holdings for each year are derived from a Fama-MacBeth model predicting cash ratios; the coefficients of which are the average coefficients from annual cross-sectional regressions estimated over the period 1980 to 1989. Estimates from this regression are as follows: Cash ratio = 0.307 + 0.230 Industry cash flow volatility + 0.006 Market to book - 0.009 Log size + 0.077 Cash flow/assets - 0.238 Net working capital/assets - 0.372 Capital expenditures/assets - 0.360 Leverage + 0.048 R&D/sales - 0.024 Dividend dummy - 0.233 Acquisition/assets + 0.158 Net equity/assets + 0.190 Net debt/assets. The table summarizes differences between actual and predicted cash ratios, by year, for the whole sample (n = 64,068 firm-year observations), for firms in the S&P 500 index (n = 5,031), and for firms paying and not paying common dividends (n = 16,206 and 47,862, respectively) during a particular year. t-statistics summarize the statistical significance of differences between predicted and actual cash ratios for the whole sample and each of the observed subsamples independently. Variable definitions are provided in the Appendix.									
Year	Whole Sample			S&P 500 Firms			Firms Paying a Dividend		
	Actual - Predicted	t-statistic	Predicted	Actual - Predicted	t-statistic	Predicted	Actual - Predicted	t-statistic	Predicted
1990	0.149	-0.015	-6.14	0.098	-0.023	-4.80	0.109	-0.011	-3.45
1991	0.170	-0.015	-6.18	0.109	-0.034	-8.78	0.120	-0.016	-4.90
1992	0.175	-0.015	-6.26	0.115	-0.039	-9.16	0.127	-0.022	-7.25
1993	0.185	-0.015	-6.49	0.120	-0.045	-10.54	0.133	-0.025	-8.08
1994	0.176	-0.024	-11.28	0.117	-0.049	-11.95	0.125	-0.035	-12.69
1995	0.180	-0.018	-7.97	0.117	-0.047	-10.56	0.122	-0.031	-10.44
1996	0.192	-0.008	-3.47	0.119	-0.047	-10.18	0.124	-0.030	-10.36
1997	0.185	-0.004	-1.60	0.120	-0.048	-10.81	0.127	-0.025	-7.48
1998	0.170	-0.004	-1.54	0.117	-0.045	-9.42	0.114	-0.029	-8.80
1999	0.189	-0.010	-4.05	0.125	-0.042	-7.42	0.114	-0.031	-8.83
2000	0.201	-0.008	-3.10	0.106	-0.054	-5.46	0.117	-0.038	-11.10
2001	0.200	-0.002	-0.72	0.144	-0.020	-2.97	0.122	-0.034	-8.92
2002	0.200	-0.002	-0.59	0.147	-0.005	-0.72	0.131	-0.006	-9.10
2003	0.212	0.001	0.38	0.153	-0.003	-0.39	0.142	-0.021	-5.65
2004	0.222	0.007	2.27	0.155	-0.006	-0.77	0.144	-0.021	-5.08
2005	0.217	0.009	2.81	0.153	-0.013	-1.74	0.145	-0.021	-5.14
2006	0.209	0.010	3.21	0.144	-0.022	-3.56	0.135	-0.023	-6.27

Table V Determinants of Changes in Predicted Cash between 2000 and 2006									
This table summarizes the determinants of the change in predicted cash ratios between 2000 and 2006, where the change in the cash ratio is measured as the difference between the average cash ratio from 2000 through 2006 and the average cash ratio from 1990 through 1989. The determinants of the cash ratio are modeled as Cash ratio = 0.307 + 0.230 Industry cash flow volatility + 0.006 Market to book - 0.009 Log size + 0.077 Cash flow/assets - 0.238 Net working capital/assets - 0.372 Capital expenditures/assets - 0.360 Leverage + 0.048 R&D/sales - 0.024 Dividend - 0.233 Acquisition/assets + 0.158 Net equity/assets + 0.190 Net debt/assets. Estimated standard errors (in parentheses) are calculated based on the Delta method discussed in Greene (1995). Variable definitions are provided in the Appendix.									
Industry sigmas	Whole Sample		Nondividend Paying Firms		Dividend Paying Firms		Difference		Difference
	Estimate	t-statistic	Estimate	t-statistic	Estimate	t-statistic	Estimate	t-statistic	
Market to book	0.002 (0.0001)	-0.008 (0.0001)	0.021 (0.0012)	0.014 (0.0008)	0.007 (0.0002)	0.001 (0.0001)	0.001 (0.0002)	0.001 (0.0001)	0.003 (0.0003)
Real size	-0.008 (0.0003)	-0.012 (0.0004)	-0.009 (0.0004)	-0.004 (0.0004)	-0.002 (0.0002)	-0.001 (0.0001)	-0.002 (0.0002)	-0.001 (0.0001)	-0.004 (0.0004)
Cash flow/ assets	0.002 (0.0002)	0.022 (0.0002)	0.002 (0.0002)	0.021 (0.0002)	0.001 (0.0007)	0.001 (0.0007)	0.001 (0.0002)	0.001 (0.0002)	0.001 (0.0002)
NWC/ assets	0.005 (0.0008)	0.022 (0.0008)	0.005 (0.0008)	0.021 (0.0007)	0.001 (0.0007)	0.001 (0.0007)	0.001 (0.0002)	0.001 (0.0002)	0.001 (0.0002)
Capex	0.013 (0.0003)	0.015 (0.0004)	0.011 (0.0004)	0.011 (0.0004)	0.004 (0.0004)	0.004 (0.0004)	0.004 (0.0004)	0.004 (0.0004)	0.001 (0.0004)
Leverage	0.010 (0.0007)	0.018 (0.0007)	0.009 (0.0008)	0.009 (0.0008)	0.022 (0.0002)	0.022 (0.0002)	0.022 (0.0002)	0.022 (0.0002)	0.014 (0.0002)
R&D/sales	0.010 (0.0003)	0.012 (0.0003)	0.009 (0.0003)	0.009 (0.0003)	0.012 (0.0003)	0.012 (0.0003)	0.012 (0.0003)	0.012 (0.0003)	0.010 (0.0003)
Dividend	0.004 (0.0002)	0.009 (0.0002)	0.009 (0.0002)	0.009 (0.0002)	0.000 (0.0000)	0.000 (0.0000)	0.004 (0.0002)	0.004 (0.0002)	-0.001 (0.0004)
Acquisition activity	-0.002 (0.0001)	-0.001 (0.0002)	-0.002 (0.0001)	-0.002 (0.0001)	0.001 (0.0002)	0.001 (0.0002)	0.001 (0.0001)	0.001 (0.0001)	0.001 (0.0001)
Net equity issuance	0.000 (0.0003)	-0.001 (0.0004)	-0.001 (0.0004)	-0.001 (0.0004)	-0.001 (0.0002)	-0.001 (0.0002)	-0.001 (0.0001)	-0.001 (0.0001)	0.003 (0.0003)
Net debt issuance	0.001 (0.0002)	-0.001 (0.0003)	-0.001 (0.0003)	-0.001 (0.0003)	0.003 (0.0003)	0.003 (0.0003)	0.003 (0.0003)	0.003 (0.0003)	0.000 (0.0004)
TOTAL	0.072	0.071	0.039	0.039	0.040	0.040	0.073	0.068	0.018

Table IV: Predicted Cash Ratios and Their Deviations from Actual Cash Holdings over Time

Table V: Determinants of Changes in Predicted Cash between 2000 and 2006

Read.

- Table IV shows that a model estimated using the 1980s predicts much of the later rise in cash holdings.

- Table V decomposes predicted cash growth and identifies four dominant components: lower NWC, higher cash-flow risk, lower capex, and higher R&D.

Economic message. The decomposition is the paper’s main explanatory mechanism: firms become riskier, more intangible, and less able to rely on working-capital liquidity.

11.7 Table VI: OLS Regression Results for the Market Value of the Firm

Table VI
OLS Regression Results for the Market Value of the Firm

The dependent variable for the regressions is the market value of the firm in year t , M_t . For each independent variable X , X_t is the level in year t , divided by the level of total assets in year t ; dX_t is the change in the level of X from year $t - 2$ to year t , divided by total assets in year t ($(X_t - X_{t-2})/A_t$); dX_{t+2} is the change in the level of X from year $t + 2$ to year t , divided by assets in year t ($(X_{t+2} - X_t)/A_t$). The first regression is for the full sample of firms that satisfy our prior data requirements, and for which we are able to calculate 2-year leads and lags, yielding a panel of 49,347 observations for 7,059 unique firms. The second regression imposes the additional requirement that assets exceed \$100 million in 2004, yielding a panel of 30,676 observations for 4,250 unique firms. p -values based on standard errors robust to clustering by firm and year are reported in parentheses. Variable definitions are provided in the Appendix.

Model	1	2
Variable		
Intercept	0.620 (0.000)	0.302 (0.000)
E_t	-0.505 (0.051)	3.852 (0.000)
dE_t	0.866 (0.000)	-0.392 (0.016)
dE_{t+2}	0.238 (0.018)	1.428 (0.000)
dNA_t	0.776 (0.000)	0.619 (0.000)
dNA_{t+2}	0.421 (0.000)	0.224 (0.000)
RD_t	3.012 (0.000)	3.468 (0.000)
dRD_t	1.647 (0.001)	2.097 (0.40)
dRD_{t+2}	3.953 (0.000)	4.463 (0.000)
I_t	1.622 (0.003)	0.512 (0.411)
dI_t	-4.079 (0.000)	-2.294 (0.006)
dI_{t+2}	-2.743 (0.000)	-1.440 (0.020)
D_t	9.180 (0.000)	6.592 (0.000)
dD_t	11.000 (0.000)	11.041 (0.000)
dD_{t+2}	15.538 (0.000)	12.113 (0.000)
dV_{t+2}	-0.083 (0.001)	-0.067 (0.076)
L_t	1.263 (0.000)	1.234 (0.000)
$L_t * D90s$	0.475 (0.032)	0.777 (0.000)
$L_t * D00s$	0.219 (0.455)	0.450 (0.199)
$D90s$	0.195 (0.000)	0.331 (0.000)
$D00s$	0.061 (0.147)	0.062 (0.164)
Adjusted R^2	0.293	0.402

Table VI: OLS Regression Results for the Market Value of the Firm

Read.

- Table VI estimates the market value of cash using valuation regressions with cash and decade interactions.

- The interaction terms do not support a decline in the value of cash in the 2000s.

Economic message. If agency problems were increasingly responsible for cash accumulation, the marginal value of cash should fall. The paper does not find that pattern.

12 Short summary

- U.S. industrial firms' average cash/assets more than doubles from 1980 to 2006.
- Net leverage falls sharply and becomes negative for the average firm in the mid-2000s.
- The increase is strongest among small firms, IPO cohorts, nondividend payers, loss firms, high-volatility industries, and R&D-intensive firms.
- A cash-demand model estimated on the 1980s explains much of later cash holdings.
- The most important drivers are falling noncash NWC, rising cash-flow volatility, falling capex, and rising R&D.
- Agency explanations are not supported by GIM sorts, value-of-cash tests, or excess-cash growth tests.

13 Conclusion

Bates, Kahle, and Stulz explain the rise in corporate cash as a rational response to changing firm characteristics and a riskier business environment. Firms increasingly rely on intangible investment, face higher cash-flow volatility, hold fewer inventories and receivables, and spend less on capital expenditures. These changes make internal liquidity more valuable.

The paper's most important conceptual point is that the meaning of leverage changes when firms hold large cash balances. Gross debt ratios suggest little change in leverage, while net debt shows a dramatic decline. By the end of the sample, the average U.S. industrial firm is far less levered on a net basis than standard debt/assets ratios imply.

The paper does not deny that some firms may hold too much cash because of agency problems. Instead, it shows that agency problems do not explain the aggregate rise. The dominant explanation is precautionary: firms hold more cash because they have become firms for which cash is more valuable.